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Our data set revolves around the Bone Marrow transplants data set (<https://archive.ics.uci.edu/dataset/565/bone+marrow+transplant+children>), which is a collection of medical data of children who received bone marrow transplants. The purpose of the data and our project is to study how different variables can be helpful in predicting patient recovery and long-term health after surgery. The initial step that we took was to closely examine what variables/covariates had missing values to see which we could use and/or what we wanted to do with the missing data. As seen in image (a), quite a few columns had missing values (RecipientABO, RecipientRh, ABOMatch, CMVstatus, DonorCMV, RecipientCMV, etc..). In this specific case of trying to predict the patient's recovery after surgery using columns with missing data, we can either remove the observations with missing data to not skew the results, or we can remove the columns themselves and try and predict using the remaining columns; this is something that we are still exploring. Another aspect we have explored is understanding each variable itself, as such we have listed below each variable and its respective description, this could help weigh in on the decision as to which variables are actually important from the ones that we are missing data from. We also created some visualizations to see the importance of each variable on the predictor as seen in (d) below.

Variable Explanations

Recipientgender: Male - 1, Female - 0

Stemcellsource: Source of hematopoietic stem cells (Peripheral blood - 1, Bone marrow - 0)

Donorage: Age of the donor at the time of hematopoietic stem cells apheresis

Donorage35: Donor age <35 - 0, Donor age >=35 - 1

IIIV: Development of acute graft versus host disease stage II or III or IV (Yes - 1, No - 0)

Gendermatch: Compatibility of the donor and recipient according to their gender (Female to Male - 1, Other - 0)

DonorABO: ABO blood group of the donor of hematopoietic stem cells (0 - 0, 1, A, B=-1, AB=2)

RecipientABO: ABO blood group of the recipient of hematopoietic stem cells (0 - 0, 1, A, B=-1, AB=2)

RecipientRh: Presence of the Rh factor on recipient's red blood cells ('+' - 1, '-' - 0)

ABOMatch: Compatibility of the donor and the recipient of hematopoietic stem cells according to ABO blood group (matched - 1, mismatched - 1)

CMVstatus: Serological compatibility of the donor and the recipient of hematopoietic stem cells according to cytomegalovirus

DonorCMV: Presence of cytomegalovirus infection in the donor of hematopoietic stem cells prior to transplantation (present, absent)

RecipientCMV: Presence of cytomegalovirus infection in the donor of hematopoietic stem cells prior to transplantation (presence - 1, absence - 0)

Disease: Type of disease (ALL,AML,chronic,nonmalignant,lymphoma)

Riskgroup: High risk - 1, Low risk - 0

Txpostrelapse: The second bone marrow transplantation after relapse (No - 0; Yes - 1)

Diseasegroup: Type of disease (malignant - 1, nonmalignant - 0)

HLAmatch: Compatibility of antigens of the main histocompatibility complex of the donor and the recipient of hematopoietic stem cells according to ALL international BFM SCT 2008 criteria (10/10 - 0, 9/10 - 1, 8/10 - 2, 7/10 - 3 (allele/antigens))

HLAmismatch: HLA matched - 0, HL mismatched - 1

Antigen: In how many antigens there is difference between the donor and the recipient (-1 - no differences, 0 - one difference, 1 (2) - two (three) differences)

Allele: In how many allele there is difference between the donor and the recipient {-1 no differences, 0 - one difference, 1 (2) (3) - two, (three, four) differences)

HLAgrl: The difference type between the donor and the recipient (HLA matched - 0, the difference is in only one antigen - 1, the difference is only in one allele - 2, the difference is only in DRB1 cell - 3, two differences (two allele or two antigens) - 4, two differences (two allele or two antigens) - 5)

Recipientage: Age of the recipient of hematopoietic stem cells at the time of transplantation

Recipientage10: Recipient age <10 - 0, Recipient age ≥10 - 1

Recipientageint: Recipient age in (0,5] - 0, (5, 10] - 1, (10, 20] - 2

Relapse: Reoccurrence of the disease (No - 0, Yes - 1)

aGvHDIIIIV: Development of acute graft versus host disease stage III or IV (Yes - 0, No - 1)

extcGvHD: Development of extensive chronic graft versus host disease (Yes - 0, No - 1)

CD34kgx10d6: CD34+ cell dose per kg of recipient body weight

CD3dCD34: CD3+ cell to CD34+ cell ratio

CD3dkgx10d8: CD3+ cell dose per kg of recipient body weight

Rbodymass: Body mass of the recipient of hematopoietic stem cells at the time of transplantation

ANCRecovery: Time to neutrophils recovery defined as neutrophils count >0.5 x 10⁹/L

PLTRecovery: Time to platelet recovery defined as platelet count >50000/mm³

time_to_aGvHD_III_IV: Time to development of acute graft versus host disease stage III or IV

survival_time: Time of observation (if alive) or time to event (if dead) in days

survival_status: Survival status (0 - alive, 1 - dead)

(a) Missing Values

```
Total Missing Values per Column:
Recipientgender      0
Stemcellsource       0
Donorage             0
Donorage35           0
IIIV                 0
Gendermatch          0
DonorABO             0
RecipientABO         1
RecipientRh          2
ABOmatch             1
CMVstatus            16
DonorCMV             2
RecipientCMV         14
Disease              0
Riskgroup            0
Txpostrelapse        0
Diseasegroup         0
HLAmatch             0
HLAmismatch          0
Antigen              1
Allele               1
HLAgrI               0
Recipientage         0
Recipientage10       0
Recipientageint      0
Relapse              0
aGvHDIIIIV           0
extcGvHD             31
CD34kgx10d6          0
CD3dCD34             5
CD3dkgx10d8          5
Rbodymass            2
ANCrecovery          0
PLTrecovery          0
time_to_aGvHD_III_IV 0
survival_time        0
dtype: int64
```

12 attributes have missing values and if used must be accounted for

```
1  from ucimlrepo import fetch_ucirepo
2
3  bone_marrow_transplant_children = fetch_ucirepo(id=565)
4
5  X = bone_marrow_transplant_children.data.features
6  y = bone_marrow_transplant_children.data.targets
7
8  missing_values_sum = X.isnull().sum()
9
10 print("Total Missing Values per Column:")
11 print(missing_values_sum)
```

(b) Summary Statistics

```
> dim(data)
[1] 187 37
```

There are 187 rows and 37 attributes in the dataset

(c) Basic statistics on each variable, and number of people who fall into each subcategory

```
> summary(data)
Recipientgender  stemcellsource  Donorage  Donorage35  IIIV  Gendermatch  DonorABO  RecipientABO  RecipientRh
0: 75           0: 42           Min.   :18.65  0:104       0: 75  0:155       -1:28      -1 :50       0 : 27
1:112           1:145           1st Qu.:27.04  1: 83       1:112  1: 32       0 :73       0 :48       1 :158
                                           Median :33.55                                1 :71       1 :75      NA's: 2
                                           Mean   :33.47                                2 :15       2 :13
                                           3rd Qu.:40.12                                NA's: 1
                                           Max.   :55.55

ABOmatch  CMVstatus  DonorCMV  RecipientCMV  Disease  Riskgroup  Txpostrelapse  Diseasegroup  HLAmatch
0 : 52     0 :48    0 :113    0 : 73      ALL      0:118      0:164          0: 32         0:94
1 :134     1 :27    1 : 72    1 :100     AML      1: 69      1: 23          1:155         1:65
NA's: 1     2 :57   NA's: 2    NA's: 14    chronic  :45
                                           lymphoma : 9
                                           nonmalignant:32
                                           3: 5

HLAmismatch  Antigen  Alel  HLAgrI  Recipientage  Recipientage10  Recipientageint  Relapse  aGVHDIIIIV  extcGVHD
0:159        -1 :93   -1 :93  0:94    Min.   : 0.600  0:99           0:47           0:159    0: 40       0 : 28
1: 28         0 :21    0 :54  1:42    1st Qu.: 5.050  1:88           1:51           1: 28    1:147       1 :128
                                           1 :65    2:14    Median : 9.600
                                           2 : 7     3: 9    Mean   : 9.932
                                           NA's: 1   3 : 1   4:19    3rd Qu.:14.050
                                           NA's: 1   5: 4    Max.   :20.200
                                           7: 5

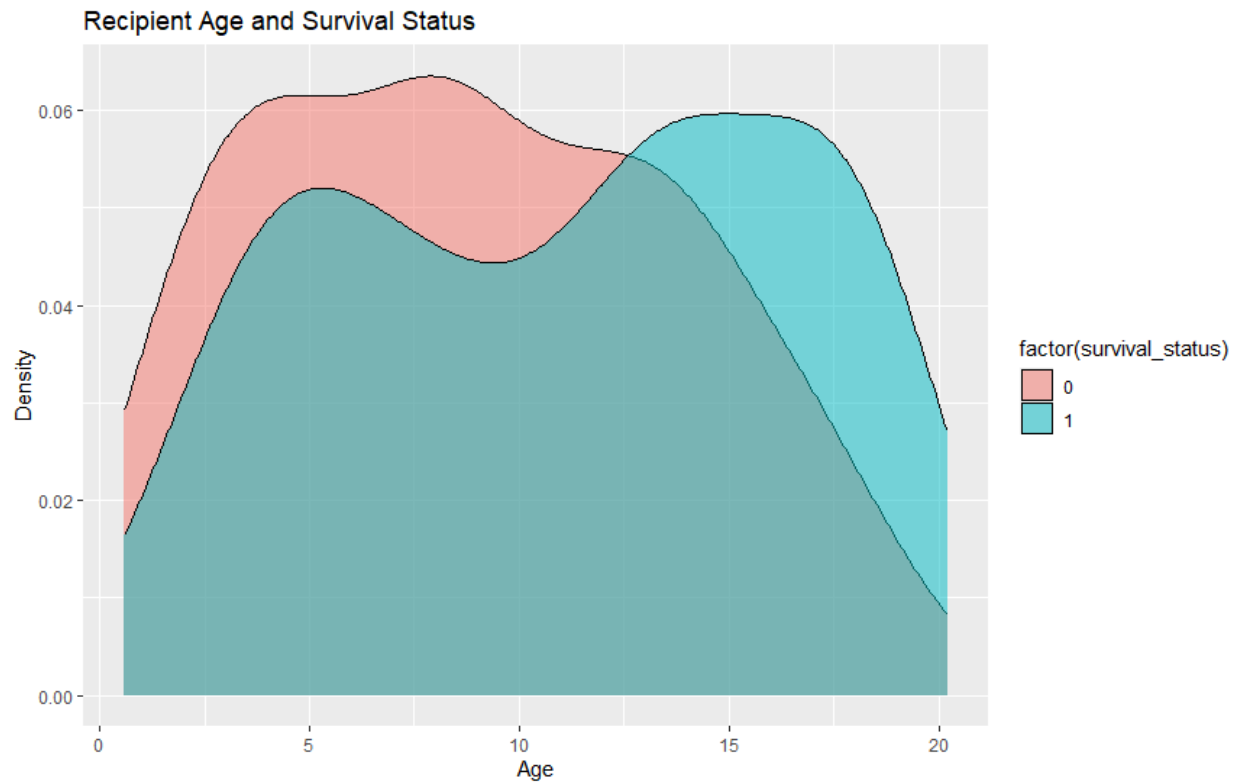
CD34kgx10d6  CD3dCD34  CD3dkgx10d8  Rbodymass  ANCrecovery  PLTcrecovery
Min.   : 0.79  Min.   : 0.2041  Min.   : 0.040  Min.   : 6.0  Min.   : 9  Min.   : 9
1st Qu.: 5.35  1st Qu.: 1.7867  1st Qu.: 1.688  1st Qu.: 19.0  1st Qu.: 13  1st Qu.: 16
Median : 9.72  Median : 2.7345  Median : 4.325  Median : 33.0  Median : 15  Median : 21
Mean   :11.89  Mean   : 5.3851  Mean   : 4.746  Mean   : 35.8  Mean   : 26753  Mean   : 90938
3rd Qu.:15.41  3rd Qu.: 5.8236  3rd Qu.: 6.785  3rd Qu.: 50.6  3rd Qu.: 17  3rd Qu.: 37
Max.   :57.78  Max.   :99.5610  Max.   :20.020  Max.   :103.4  Max.   :1000000  Max.   :1000000
                                           NA's :5      NA's :5      NA's :2

time_to_agvhd_III_IV  survival_time  survival_status
Min.   : 10        Min.   : 6.0    Min.   :0.0000
1st Qu.:1000000    1st Qu.: 168.5  1st Qu.:0.0000
Median :1000000    Median : 676.0  Median :0.0000
Mean   : 775408    Mean   : 938.7  Mean   :0.4545
3rd Qu.:1000000    3rd Qu.:1604.0  3rd Qu.:1.0000
Max.   :1000000    Max.   :3364.0  Max.   :1.0000
```

(d) EDA Graphs

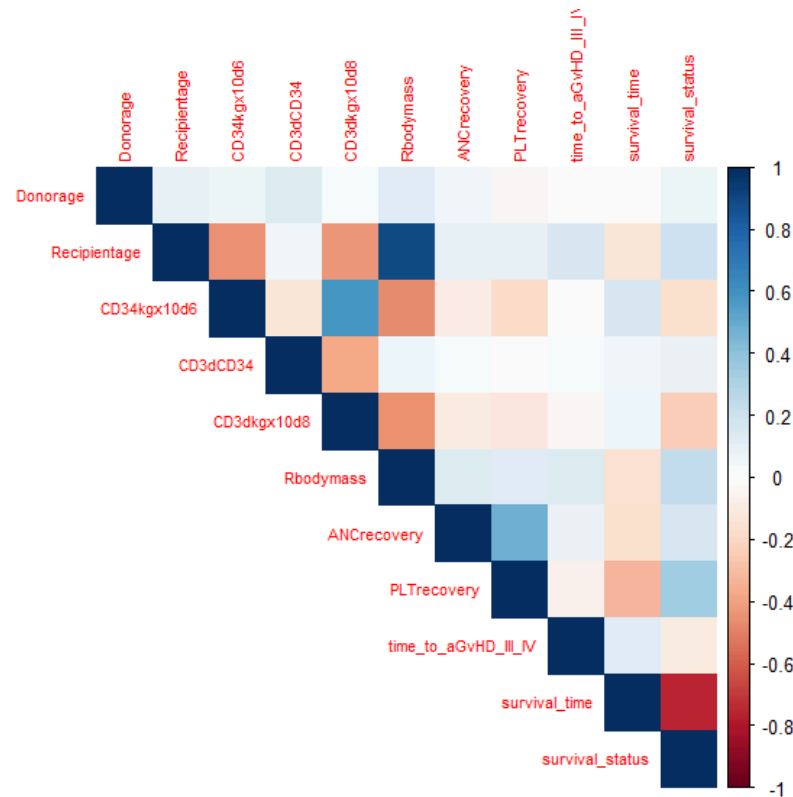
(Code for age/survival status)

```
> ggplot(data, aes(x = Recipientage, fill = factor(survival_status))) +  
+   geom_density(alpha = 0.5) +  
+   labs(title = "Recipient Age and survival status", x = "Age", y = "Density")
```



(Code for correlation heatmap)

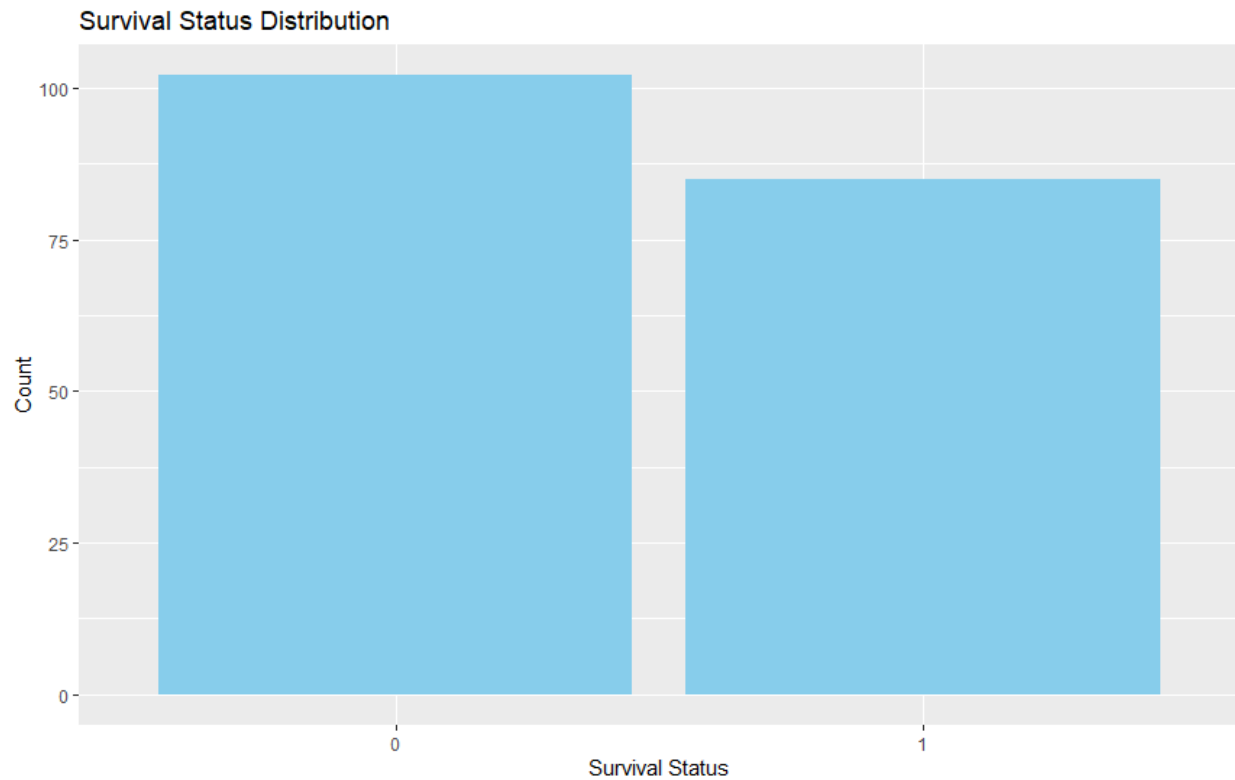
```
> numeric_data <- data[, sapply(data, is.numeric)]  
> correlation_matrix <- cor(numeric_data, use = "complete.obs")  
> corrplot(correlation_matrix, method = "color", type = "upper", tl.cex = 0.7)
```



- Survival status and survival time has a strong negative correlation which makes sense because higher survival time should correspond to patients who are still alive and vice versa.
- Rbodymass and recipientage have a moderate correlation, which indicates older patients have more body mass.
- TimetoaGvHDIIIIV has weak correlation compared to every other variable, can indicate this variable acts independently towards survival.
- Survivaltime is positively correlated with ANCrecovery and PLTrecovery.

(Code for survival status distribution)

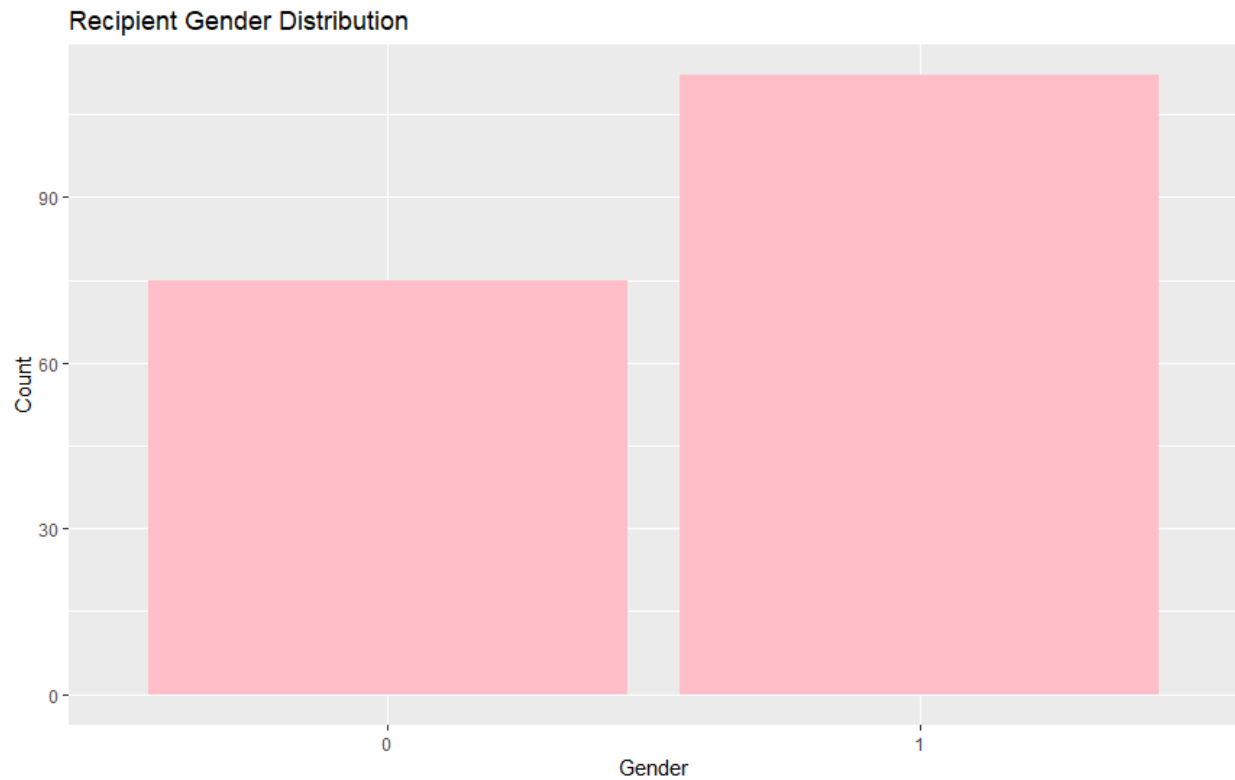
```
> ggplot(data, aes(x = factor(survival_status))) +
+   geom_bar(fill = "skyblue") +
+   labs(title = "Survival Status Distribution", x = "Survival Status", y = "Count")
```



There is an approximately equal number of each class, meaning there shouldn't be issues of uneven classes interfering with classifications

(Code for recipient gender)

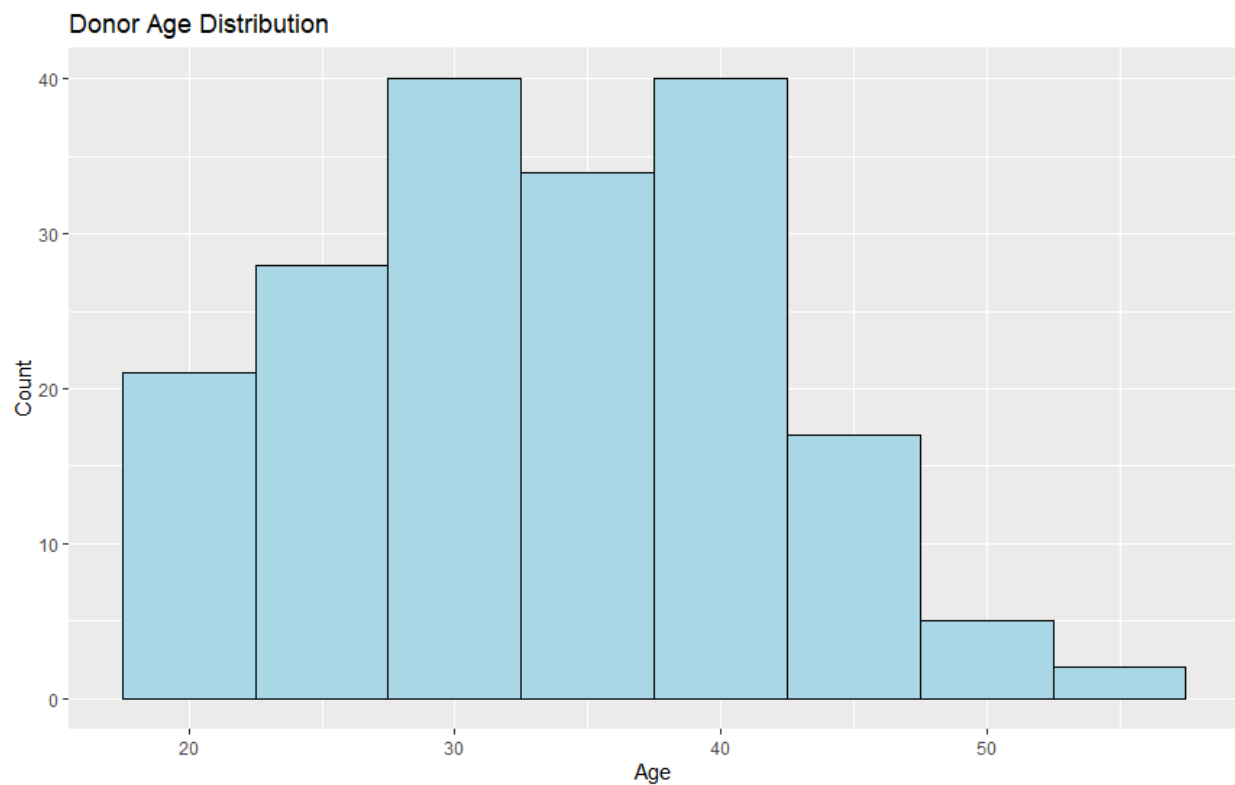
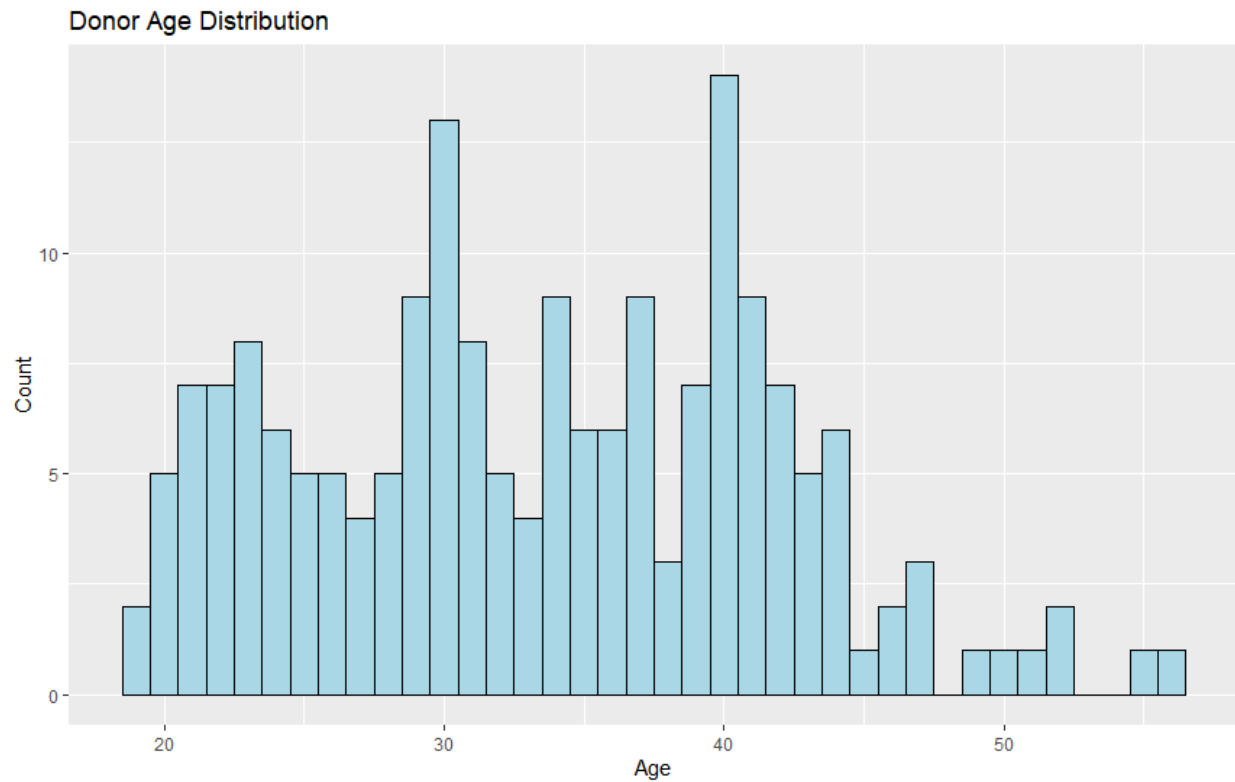
```
> ggplot(data, aes(x = factor(Recipientgender))) +  
+   geom_bar(fill = "pink") +  
+   labs(title = "Recipient Gender Distribution", x = "Gender", y = "Count")
```



There is a roughly equal amount of category

(Code for donor age distribution)

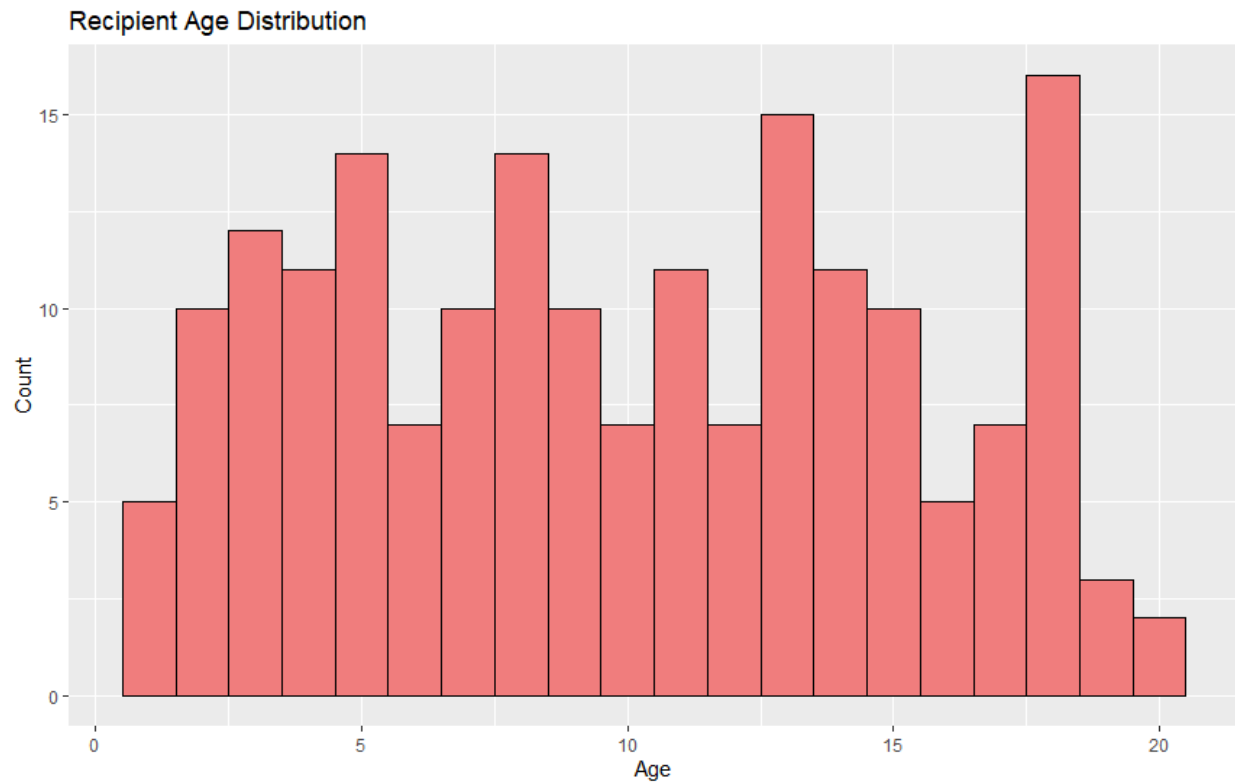
```
> ggplot(data, aes(x = Donorage)) +  
+   geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +  
+   labs(title = "Donor Age Distribution", x = "Age", y = "Count")
```

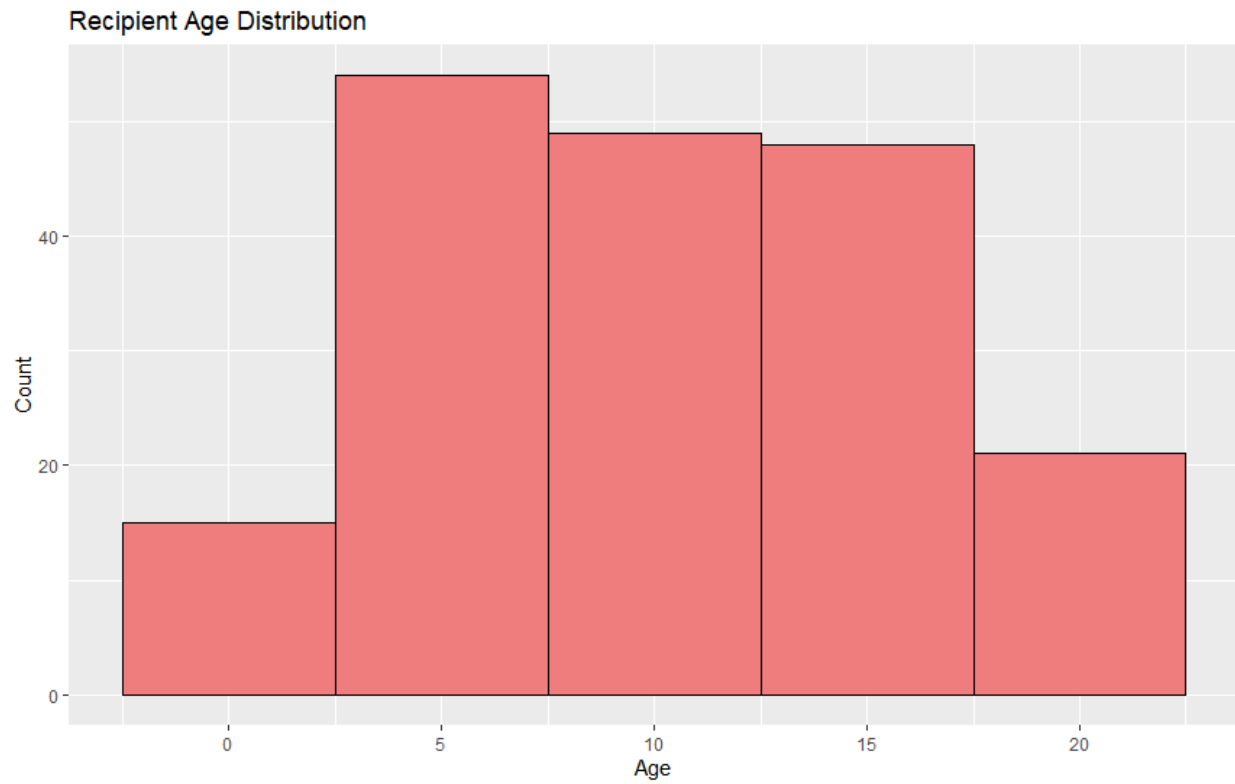


The distribution is roughly normal but does have some notable outliers that could interfere with predicting

(Code for recipient age distribution)

```
> ggplot(data, aes(x = Recipientage)) +  
+   geom_histogram(binwidth = 1, fill = "lightcoral", color = "black") +  
+   labs(title = "Recipient Age Distribution", x = "Age", y = "Count")
```





The distribution is roughly uniform but does have some notable outliers that could interfere with predicting

(e) Available Predicting Variables

Variables that are known before the transplant:

Recipientgender: Male - 1, Female - 0

Stemcellsource: Source of hematopoietic stem cells (Peripheral blood - 1, Bone marrow - 0)

Donorage: Age of the donor at the time of hematopoietic stem cells apheresis

Donorage35: Donor age <35 - 0, Donor age ≥35 - 1

Gendermatch: Compatibility of the donor and recipient according to their gender (Female to Male - 1, Other - 0)

DonorABO: ABO blood group of the donor of hematopoietic stem cells (0 - 0, 1, A, B=-1, AB=2)

RecipientABO: ABO blood group of the recipient of hematopoietic stem cells (0 - 0, 1, A, B=-1, AB=2)

RecipientRh: Presence of the Rh factor on recipient's red blood cells ('+' - 1, '-' - 0)

ABOmatch: Compatibility of the donor and the recipient of hematopoietic stem cells according to ABO blood group (matched - 1, mismatched - 1)

CMVstatus: Serological compatibility of the donor and the recipient of hematopoietic stem cells according to cytomegalovirus

DonorCMV: Presence of cytomegalovirus infection in the donor of hematopoietic stem cells prior to transplantation (present, absent)

RecipientCMV: Presence of cytomegalovirus infection in the donor of hematopoietic stem cells prior to transplantation (presence - 1, absence - 0)

Disease: Type of disease (ALL,AML,chronic,nonmalignant,lymphoma)

Riskgroup: High risk - 1, Low risk - 0

Diseasegroup: Type of disease (malignant - 1, nonmalignant - 0)

HLAmatch: Compatibility of antigens of the main histocompatibility complex of the donor and the recipient of hematopoietic stem cells according to ALL international BFM SCT 2008 criteria (10/10 - 0, 9/10 - 1, 8/10 - 2, 7/10 - 3 (allele/antigens))

HLAmismatch: HLA matched - 0, HL mismatched - 1

Antigen: In how many antigens there is difference between the donor and the recipient (-1 - no differences, 0 - one difference, 1 (2) - two (three) differences)

Allele: In how many allele there is difference between the donor and the recipient {-1 no differences, 0 - one difference, 1 (2) (3) - two, (three, four) differences)

HLAgrl: The difference type between the donor and the recipient (HLA matched - 0, the difference is in only one antigen - 1, the difference is only in one allele - 2, the difference is only in DRB1 cell - 3, two differences (two allele or two antigens) - 4, two differences (two allele or two antigens) - 5)

Recipientage: Age of the recipient of hematopoietic stem cells at the time of transplantation

Recipientage10: Recipient age <10 - 0, Recipient age ≥10 - 1

Recipientageint: Recipient age in (0,5] - 0, (5, 10] - 1, (10, 20] - 2

CD34kgx10d6: CD34+ cell dose per kg of recipient body weight

CD3dCD34: CD3+ cell to CD34+ cell ratio

CD3dkgx10d8: CD3+ cell dose per kg of recipient body weight

Rbodymass: Body mass of the recipient of hematopoietic stem cells at the time of transplantation